

Sprint 7: Cached Network Fee Engine

Fast withdrawal fee quotes from fresh network-scoped evidence

Delivered. Local quality gates and a full local PostgreSQL run pass; production migration remains a separate approval.

Honest status: ADR-012 is accepted. Domain calculations, schema, migration 12, cache repository, leased worker, strict withdrawal-fee-quote API, and automated tests are implemented and merged to `main`. Migration 12 applied successfully to a local PostgreSQL instance. All 3 integration suites (network fee, pricing, treasury; 13 tests) pass on a clean full rerun, each isolated in its own OS process so a degraded connection pool in one suite cannot affect another. GitHub Actions cannot start a runner for this repository's account (an account-level billing state affecting every workflow run, not this branch's code); see ADR-013 for the accepted local-verification substitute used to reach this status.

Connection to Earlier Sprints

Sprint 3 provides immutable conversion-rate evidence. Sprint 5 provides network-scoped treasury configuration. Sprint 6 reserves purchased inventory. Sprint 7 adds a separate cache for the blockchain cost of external withdrawals; it does not change purchase quote spreads or fees.

What Was Built

The pure domain engine selects the conservative estimate, checks provider/RPC deviation, applies fixed and percentage buffers with upward integer rounding, and converts native gas cost into the withdrawn asset using immutable rational evidence. PostgreSQL stores versioned policies, observations, conversion evidence, accepted or rejected snapshots, exact snapshot inputs, leased refresh jobs, and immutable withdrawal fee quotes. The worker uses deterministic sandbox adapters behind domain-owned interfaces. The API requires an explicit asset-network and transfer type.

End-to-End Example

```
fee provider estimate ---- independent RPC estimate
      \                               /
      freshness + deviation gate
            |
      immutable cached snapshot
            |
      native gas + buffer + conversion evidence
            |
Sendaza fee request: asset-network + type + amount
            |
principal + network charge + service fee = total debit
            |
      immutable quote with a short expiry
```

For a token withdrawal, the recipient still receives the exact principal. The treasury may pay gas in the network's native asset, while the customer network-fee charge is converted into the withdrawn asset. USDT on two networks therefore uses two independent policies and caches.

System Boundary

Sendaza owns the customer screen, authorization, and ledger lock. SLE owns the estimate, buffer, conversion evidence, and quote. Sprint 8 will consume the fee quote during Fireblocks submission. It may reject or review a withdrawal if current cost exceeds tolerance, but it cannot silently increase the approved customer debit.

Rules Protected

- Amounts use atomic-unit `bigint` ; API amounts are decimal strings.
- Fee routes are keyed by asset-network and transfer type, never asset alone.
- Stale, missing, duplicated-source, or deviating evidence cannot publish an accepted snapshot.
- Buffers, conversions, and service percentages round upward.
- Recipient principal is unchanged; fees are explicit add-ons.
- Cross-asset customer debits are rejected for the MVP.
- Evidence, snapshots, and quotes are immutable in PostgreSQL.
- Quote creation is idempotency- and audit-protected.

Test and Environment Evidence

Local gates currently cover exact arithmetic, evidence failures, strict request parsing, response precision, and controller behavior: 44 unit suites, 247 tests. Migration 12 applied cleanly to a local PostgreSQL 16 instance; Prisma reported all 12 migrations current. A full local PostgreSQL integration run passed all 3 suites (network fee, pricing, treasury) and all 13 tests, each suite executed as its own OS process via `scripts/run-integration-tests.cjs` so one suite's degraded connection pool cannot hang or fail another. GitHub Actions itself remains blocked at the account level for this repository (ADR-013); local verification is the currently accepted substitute evidence.

Limitations and Next Dependency

The active worker uses deterministic sandbox estimators. Real provider and independent RPC adapters require network-specific integration and credentials before mainnet. Sprint 7 does not submit withdrawals or record mined fees. Production migration 12 and a real Railway/CI run remain separate approvals once GitHub Actions billing is resolved for this account.

Glossary

Native fee asset: the coin a blockchain uses for gas. **Charge asset:** the asset used to recover the network cost from the customer. **Buffer:** an explicit safety amount above the estimate. **Deviation:** the difference between independent estimates. **Snapshot:** immutable cached evidence valid until a fixed time. **Rational conversion:** an exact numerator and denominator rather than floating-point math. **Lease:** a time-limited worker claim safe to recover after a crash.

Sprint 7 handover report. Canonical behavior lives in ADR-012, API_SPEC.md, ARCHITECTURE.md, and modules/network-fees/MODULE.md.